Automated Arduino Traffic Light

Objective:

In this assignment, you will build and program a fully automated traffic light system using an **Arduino Uno** micro-controller and **Tinkercad Circuits**. You will write code (using visual Blocks or C++ text) to cycle through Red, Yellow, and Green LEDs in a real-world sequence.

Name : ABDUL HARIZ B. MOHD SARNUDIN

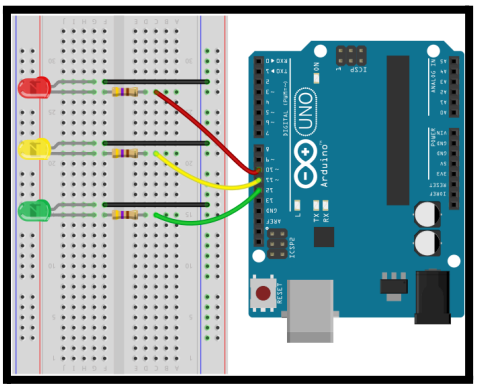
Student ID : CBS24070208

Semester/Trade/Group : 5 / DCBS / CBS 1

TTO : SIR LUQMAN HAKIM B. MOHD SABANI

Required Components (Tinkercad Circuits)

- 1x Arduino Uno R3
- 1x Small Breadboard
- 3x LEDs (1 Red, 1 Yellow, 1 Green)
- 3x 220 Resistors
- Jumper Wires



Step-by-Step Instructions

Part A: Wiring the Circuit

1. **Open Tinkercad:** Click the Tinkercad attachment in this assignment to launch the workspace.
2. **Create Circuit:** Go to **Designs** > **Create** > **Circuit**.
3. **Place Components:** Add the **Arduino Uno** and **Small Breadboard** to your workplane.
4. **Connect Ground (GND):** Wire a GND pin from the Arduino to the negative (-) rail on the breadboard.
5. **Place LEDs:** Place 3 LEDs side-by-side on the breadboard and change their colors to **Red**, **Yellow**, and **Green**.
6. **Add Resistors:** Connect a **220 Resistor** from the Cathode (short leg) of each LED to the negative (-) ground rail.
7. **Connect Digital Pins:**
    - Wire Digital **Pin 12** to the Anode (long leg) of the **Red LED**.
    - Wire Digital **Pin 11** to the Anode (long leg) of the **Yellow LED**.
    - Wire Digital **Pin 10** to the Anode (long leg) of the **Green LED**.

Part B: Programming the Traffic Light

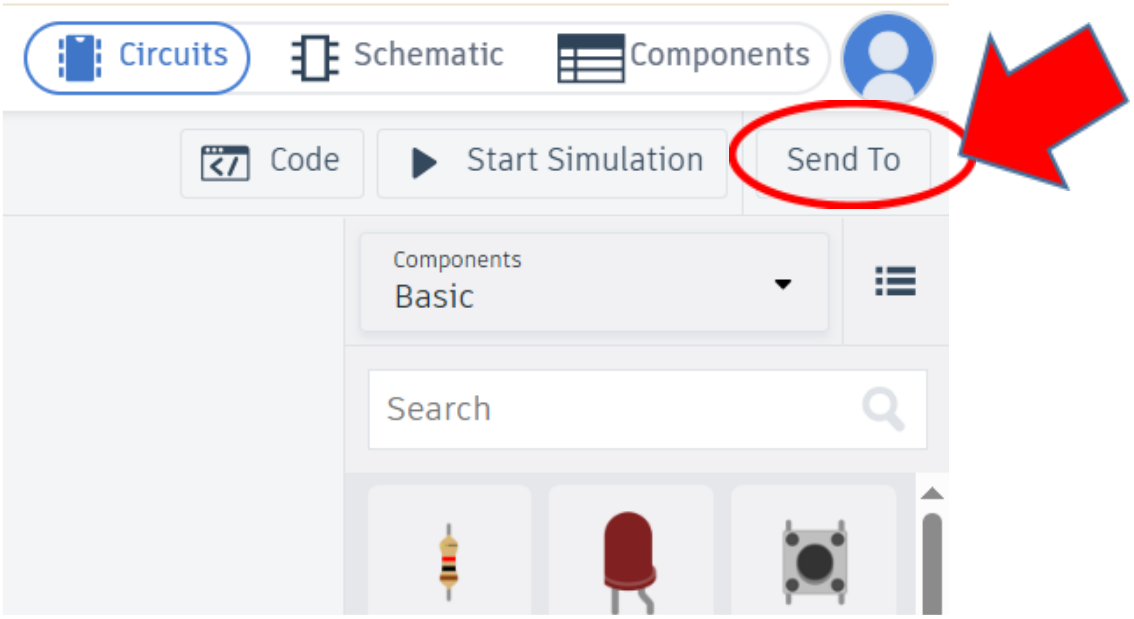
1. Click the **Code** button in the top right toolbar (set editor mode to **Blocks** or **Blocks + Text**).
2. Program the sequence using pin output blocks and delay commands:

- **Red Light:** Set Pin 12 HIGH Wait 3 seconds Set Pin 12 LOW.
- **Green Light:** Set Pin 10 HIGH Wait 3 seconds Set Pin 10 LOW.
- **Yellow Light:** Set Pin 11 HIGH Wait 1 second Set Pin 11 LOW.

3. Click **Start Simulation** to test if your lights cycle continuously in the correct traffic order.

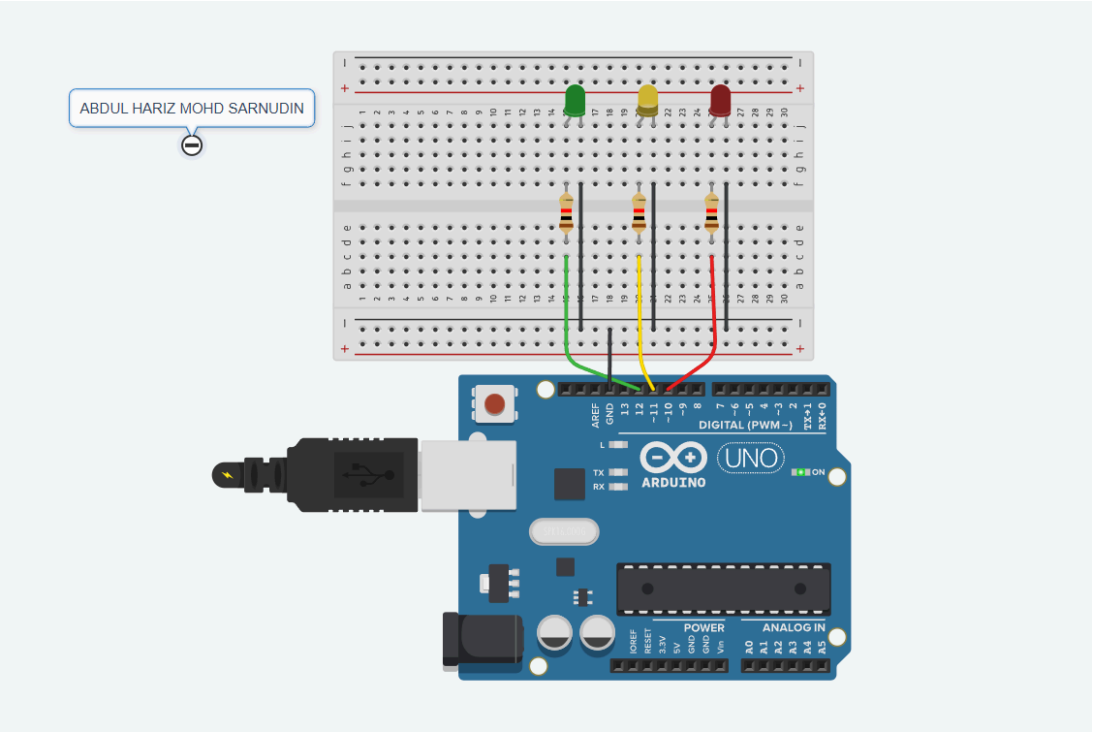
Part C: Submission

- Click **Send To** (top right corner) > **Share to Google Classroom** to turn in your project.



Copy and paste your diagram:

Example:



Code | Stop Simulation | Send To

1 (Arduino Uno R3)

```
// Define the Arduino pin numbers
const int RED_LED = 10;
const int YELLOW_LED = 11;
const int GREEN_LED = 12;

void setup() {
  // Configure the LED pins as outputs
  pinMode(RED_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
  pinMode(GREEN_LED, OUTPUT);
}

void loop() {
  // 1. Turn Red LED ON for 1 second
  digitalWrite(RED_LED, HIGH);
  delay(1000);
  digitalWrite(RED_LED, LOW);

  // 2. Turn Yellow LED ON for 1 second
  digitalWrite(YELLOW_LED, HIGH);
  delay(1000);
  digitalWrite(YELLOW_LED, LOW);

  // 3. Turn Green LED ON for 1 second
  digitalWrite(GREEN_LED, HIGH);
  delay(1000);
  digitalWrite(GREEN_LED, LOW);
}
```